



QR Code for
report verification

Case ID	: 2600135670	Sample Type	: FFPE TISSUE BLOCK
Name	: MS. USHABAI PUNDALIK BARI (OQG2604290894)	Date & Time Collected	: 29-Apr-2026 12:00 AM
Sex/Age	: Female/54 Years	Date & Time Received	: 01-May-2026 06:23 PM
Bill. Loc.	: OncQuest Laboratories Ltd., New Delhi	Date & Time Reported	: 12-May-2026 06:36 PM
Ref. By	:	Report Version	: 1
Indication	:		

Endometseq Genomic Profiling Panel- Basic
On Illumina Novaseq 6000 Platform

Clinical Indication:

Moderately-differentiated endometrioid adenocarcinoma (Grade 2) of the endometrium.

Block No.: AH-536/26-D

Positive

(Clinically significant Variant detected)

Report Highlights:

Tier I	Tier II	Tier III	MSI	Therapies	Clinical Trials
0	1	0	Stable	1	14

Key Findings:

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
IIC	PTEN NM_000314.8	c.46_47insT p.(Y16Lfs*28)	1	483x / 3.93%

*Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines.

**Public data sources included in relevant targeted therapies, guidelines: EMA, FDA, ESMO, NCCN.

Multi-Gene Biomarkers Results:

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected

Correlation of the genetic findings with the clinical condition of the patient is required to arrive at accurate diagnosis, prognosis or therapeutic decisions.

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Clinical Significance:

Variant	Therapy	
	In This Cancer	In Other Cancer
IIC PTEN p.(Y16Lfs*28)	▪ None*	▪ capivasertib + hormone therapy

Variant	Diagnostic Significance	Prognostic Significance
PTEN p.(Y16Lfs*28)	None*	None*

Biomarker Description:

Variant	Biomarker Description
PTEN NM_000314.8 chr10:89624271 c.46_47insT p.(Y16Lfs*28)	The PTEN gene encodes the phosphatase and tensin homolog, a tumor suppressor protein with lipid and protein phosphatase activities[1]. PTEN antagonizes PI3K/AKT signaling by catalyzing the dephosphorylation of phosphatidylinositol (3,4,5)-trisphosphate (PIP3) to PIP2 at the cell membrane, which inhibits the activation of AKT[2,3]. In addition, PTEN has been proposed to influence RAD51 loading at double strand breaks during homologous recombination repair (HRR) and regulate the G2/M checkpoint by influencing CHEK1 localization through AKT inhibition, thereby regulating HRR efficiency[4]. Germline mutations in PTEN are linked to hamartoma tumor syndromes, including Cowden disease, which are defined by uncontrolled cell growth and benign or malignant tumor formation[5]. PTEN germline mutations are also associated with inherited cancer risk in several cancer types[6]. PTEN is frequently altered in cancer by inactivating loss-of-function mutations and by gene deletion. PTEN mutations are frequently observed in 50%-60% of uterine cancer[7,8]. Nearly half of somatic mutations in PTEN are stop-gain or frame-shift mutations that result in truncation of the protein reading frame. Recurrent missense or stop-gain mutations at codons R130, R173, and R233 result in loss of phosphatase activity and inhibition of wild-type PTEN[3,9,10,11,12]. PTEN gene deletion is observed in 15% of prostate cancer, 9% of squamous lung cancer, 9% of glioblastoma, and 1-5% of melanoma, sarcoma, and ovarian cancer[7,8]. Due to the role of PTEN in HRR, poly(ADP-ribose) polymerase inhibitors (PARPi) are being explored as a potential therapeutic strategy in PTEN deficient tumors[13,14]. In 2022, the FDA granted fast track designation to the small molecule inhibitor, pidnarulex[15], for BRCA1/2, PALB2, or other

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homologous recombination deficiency (HRD) mutations in breast and ovarian cancers. In 2023, the FDA approved the kinase inhibitor, capivasertib[16] in combination with fulvestrant for locally advanced or metastatic hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative breast cancer with one or more PIK3CA/AKT1/PTEN-alterations following progression after endocrine treatment.

capivasertib + fulvestrant
with ERBB2 negative, Hormone receptor positive
TRUQAP® is a kinase inhibitor indicated, in combination with fulvestrant for the treatment of adult patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, locally advanced or metastatic breast cancer with one or more PIK3CA/AKT1/PTEN-alterations as detected by an FDA-approved test following progression on at least one endocrine-based regimen in the metastatic setting or recurrence on or within 12 months of completing adjuvant therapy.
Cancer type: Breast Cancer

Therapy & Clinical Trials:

PTEN p.(Y16Lfs*28) c.46_47insT

Therapy	FDA	NCCN	EMA	ESMO	Trials
capivasertib + fulvestrant	✓	✓	✓	X	
atezolizumab	X	X	X	X	II
azenosertib	X	X	X	X	II
niraparib	X	X	X	X	II
ART-6043, olaparib, niraparib	X	X	X	X	I/II
atezolizumab, ipatasertib	X	X	X	X	I/II
ATRN-119	X	X	X	X	I/II
niraparib, GSK-101	X	X	X	X	I/II
VB-15010	X	X	X	X	I/II
HRS-1167	X	X	X	X	I

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HS-10502	X	X	X	X	I
IPN-60090, capivasertib	X	X	X	X	I
SIM-0501	X	X	X	X	I
talazoparib, palbociclib, axitinib, crizotinib	X	X	X	X	I
XL-309, olaparib	X	X	X	X	I

☑ Note: ✓ : In this Cancer Type, ✓ : In Other Cancer Type, ✓✓ : both In this and Other Cancer Type, X : No Evidence. ⚠ : Caution for Use

Test Description

Endometseq Genomic Profiling Panel – Basic is advancing precision oncology research through the analysis of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Endometseq Genomic Profiling Panel – Basic assay analyses **43** unique genes (**43** DNA) for single gene and multiple gene biomarker insights, detects single gene biomarkers from all variant types including **SNVs**, **indels**, and **CNVs**, analyze complex multi-gene biomarkers for mutational signatures, including microsatellite instability (**MSI**) for immunotherapy research.

The performance of the panel was verified using both commercially sourced reference controls and FFPE samples.

Test Methodology

Extraction & Library Preparation: DNA extracted from FFPE tissue block sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

Sequencing: The libraries were sequenced at mean depth: >1000x on Illumina Novaseq- next generation sequencing platform.

Data Analysis: Sequencing data are analyzed using a customized and validated bioinformatics pipeline capable of detecting various classes of genomic alterations, including single nucleotide variants (**SNVs**), **Indels**, and copy number variations (**CNVs**). As part of the streamlined bioinformatics workflow, Multi-gene Biomarkers such as Microsatellite Instability (**MSI**) Score is calculated. Reportable genomic alterations are

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prioritized, classified, and reported based on AMP-ASCO-CAP guidelines and NCCN guidelines.
Limit of Detection (LOD): The LOD for SNVs and InDels is 5% Variant allele Frequency (VAF).

Notes

Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions.

Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported.

SN Genelab Pvt Ltd does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently.

Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently.

All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence.

This test is performed on tumor tissue without a matched normal (e.g., blood) sample. Therefore, some variants may be of germline origin. While the assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

FFPE-derived samples often pose challenges due to degradation. Poor-quality DNA or RNA may fail QC thresholds at various stages such as extraction, library preparation, or bioinformatics, and may result in

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assay failure or compromised results, such as low gene coverage or shallow variant depth.

Variants with allele frequencies between 3–5% may be reported if they are of strong or potential clinical relevance, or if previously detected in the same patient. False positives or negatives below assay sensitivity cannot be ruled out.

CNVs/gene amplifications have >80% detection sensitivity in this assay. Confirmation using orthogonal methods such as FISH/DDPCR is recommended when clinically indicated.

Variants in non-coding regions or deep intronic sequences are not assessed in this assay.

This assay is under the scope of NABL as Solidseq Comprehensive panel.

DNA Gene List

AKT1	ARID1A	ARID5B	ATM	BCOR
BRAF	BRCA1	BRCA2	CCNE1	CDKN2A
CHEK2	CTCF	CTNNB1	EGFR	EPCAM
ERBB2	FBXW7	FGFR1	FGFR2	FGFR3
KMT2D	KRAS	MLH1	MLH3	MSH2
MSH3	MSH6	MUTYH	NTRK1	NTRK2
NTRK3	PIK3CA	PIK3R1	PIK3R2	PMS1
PMS2	POLD1	POLE	PPP2R1A	PTEN
RET	TP53	TSC2		

MSI

% of Unstable Loci

MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>30%

Tier

Interpretation

Tier IA	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to

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	a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies.
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association

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----- End Of Report -----

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